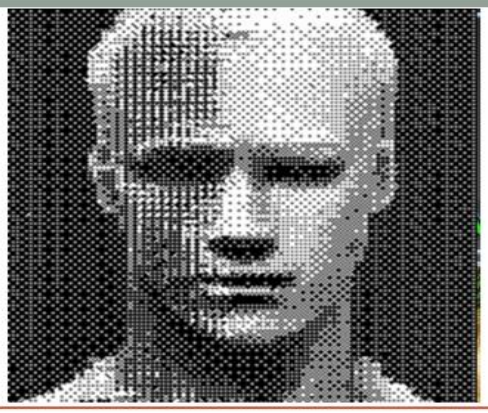


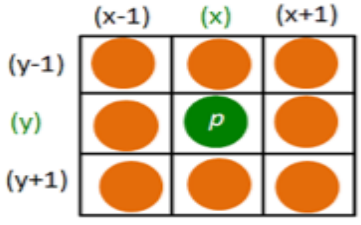
DIGITAL IMAGE PROCESSING

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Neighborhood operations processing

- The enhancement methods discussed so far are pixels transformation and does not consider the pixels neighbors.
- If there is a black pixel in a totally white region so it's definitely a noise that makes it black.
- In order to detect and remove such noise we need neighbor pixels.



- A lot of things can be achieved by neighborhood processing not possible by point processing such as.
  - Smoothing
  - Salt and pepper noise removal while preserving edges
  - Edge detection

Smoothing Spatial filters

Types of Smoothing Filters:

Averaging Filter:

- It average the pixel values within a neighborhood.
- The size of the filter kernel affects the degree of smoothing where larger kernels yields to more smoothing but also more edge blurring.

$\frac{1}{9} \times$

1	1	1
1	1	1
1	1	1

averaging filter

Weighted Averaging Filter:

- Assigns different weights to the pixels in the neighborhood, meaning some pixels contribute more to the average than others. Where the weights can be based on various criteria (e.g., distance from the center pixel, Gaussian distribution).
- It provide more flexible than the averaging filter, allowing for better edge preservation while still reducing noise.

$\frac{1}{16} \times$

1	2	1
2	4	2
1	2	1

weighted averaging

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Smoothing Spatial filters

Linear Low pass Filters kernels:

ODD# (rows) × ODD# (columns)

$\frac{1}{16} *$

1	2	1
2	4	2
1	2	1

3*3 Gaussian Kernel

$\frac{1}{9} *$

1	1	1
1	1	1
1	1	1

3*3 Averaging Kernel

$\frac{1}{21} *$

0	1	1	1	0
1	1	1	1	1
1	1	1	1	1
1	1	1	1	1
0	1	1	1	0

5*5 Circular Kernel

$\frac{1}{81} *$

1	2	3	2	1
2	4	6	4	2
3	6	9	6	3
2	4	6	4	2
1	2	3	2	1

5*5 Pyramidal Kernel

$\frac{1}{25} *$

0	0	1	0	0
0	2	2	2	0
1	2	5	2	1
0	2	2	2	0
0	0	1	0	0

5*5 Cone Kernel

Smoothing Spatial filters

Border pixels Treatments for filters:

- Ignore

  No processing required for those pixels.

- Reflection column and Row

  The matrix rows and columns are extended out with duplication to the values of the borders.

- Null values

  The matrix rows and columns are extended out with zeros.

Smoothing Spatial filters

For the following image matrix apply 3*3 averaging **filter ignoring padding and comment on the result**?

```
[255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255,   2 , 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255, 255, 255, 255, 255]
```

Smoothing Spatial filters

$\frac{1}{9}x$

x1	x1	x1
x1	x1	x1
x1	x1	x1

```
[255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255,   2 , 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255, 255, 255, 255, 255]
```

Answer

This process is repeated for every pixel in the original image to generate the smoothed image

255	255	255	255	255	255	255	255
255	255	255	255	255	255	255	255
255	255	255	227	198	170	198	255
255	255	255	198	142	85	142	255
255	255	255	170	85	0	85	255
255	227	227	170	85	0	85	255
255	227	227	198	142	85	142	255
255	255	255	255	255	255	255	255

Smoothing Spatial filter (Order-Statistics Filter)

```
[120,  105,  117,  100;
 120,  125,  100,  130;
 115,  107,  112,  118;
 130,  140,  145,  151]
```

Input image

Common types of Order-Statistics Filter:

- **Median Filter:**
  - The new value of a pixel is replaced by the median (middle) ranking value of the pixels in its neighborhood, which makes it effective for **removing salt-and-pepper noise** while preserving edges.

100, 105, 107, 112, 115, 117, 120, 120, 125

115

- **Minimum Filter**
  - The new value of a pixel is replaced by the minimum (first) ranking value of the pixels in its neighborhood, which makes it effective for **suppressing bright spots (noise) by finding the darkest points** in images.
- **Maximum Filter** **100**
  - The new value of a pixel is replaced the maximum (Last) ranking value of the pixels in its neighborhood, which makes it Effective for **suppressing dark spots (noise) by finding the brightest points** in images.

125

Sharpening Spatial filters

- **Sharpening filters:**
  - It **increase the visibility of edges** by finding changes between pixels' values.
  - Termed as a derivative filter as it produces values only if there is a change.
  - The **filter kernel must satisfy the zero-sum Property** (The sum of the kernel coefficients is zero) to ensure that the filter finds differences between pixel values.
  - Disadvantages: the sensitivity to noise (increases the spark noise), so it needs a smooth filter before detecting edges.

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Sharpening Spatial filters

Widely used sharpening filters are:

- Sobel operators

detect horizontal and vertical edges by using two separate kernels. It is effective for edge detection in specific orientations.

−1	−2	−1
0	0	0
1	2	1

Gy

−1	0	1
−2	0	2
−1	0	1

Gx

$$\text{mag} = \sqrt{Gx^2 + Gy^2}$$

- Laplacian operator

It captures edges in all directions equally and can be more sensitive to noise because it does not incorporate any smoothing.

1	1	1
1	−8	1
1	1	1

Sharpening Spatial filters

For the following image matrix apply 3*3 Sobel filter in horizontal direction **ignoring padding and comment on the result**?

```
[255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255,   2 , 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255, 255, 255, 255, 255]
```


Smoothing Spatial filters

The resultant smoothed image after normalization process

$$output_{norm} = \frac{output - output_{min}}{output_{max} - output_{min}} * 255$$

255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	255
255	0	0	255	765	1020	765	255
255	0	0	255	765	1020	765	255
255	0	0	0	0	0	0	255
255	506	253	0	0	0	0	255
255	0	0	255	765	1020	765	255
255	255	255	255	255	255	255	255

Normalization

255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	255
255	0	0	64	191	255	191	255
255	0	0	64	191	255	191	255
255	0	0	0	0	0	0	255
255	127	63	0	0	0	0	255
255	0	0	64	191	255	191	255
255	255	255	255	255	255	255	255

Threshold

255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	255
255	0	0	255	255	255	255	255
255	0	0	255	255	255	255	255
255	0	0	0	0	0	0	255
255	255	253	0	0	0	0	255
255	0	0	255	255	255	255	255
255	255	255	255	255	255	255	255

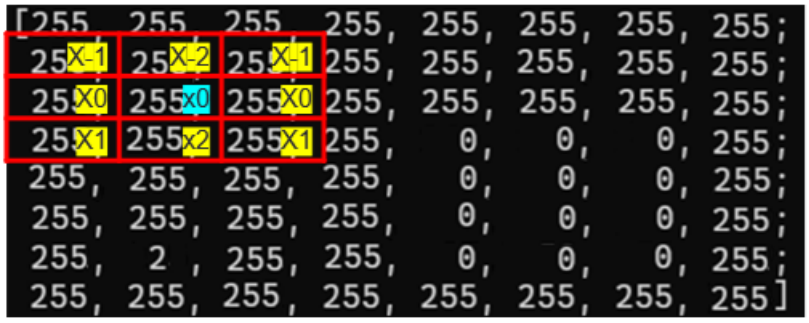
The resultant smoothed image after thresholding process

Sharpening Spatial filters

For the following image matrix apply 3*3 Sobel filter in both horizontal and vertical direction **ignoring padding and comment on the result**?

```
[255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255, 255, 255, 255, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255,   0,   0,   0, 255;
 255,   2 , 255, 255,   0,   0,   0, 255;
 255, 255, 255, 255, 255, 255, 255, 255]
```


Smoothing Spatial filters



Answer:

Gy

255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	255
255	0	0	-255	-765	-1020	-765	255
255	0	0	-255	-765	-1020	-765	255
255	0	0	0	0	0	0	255
255	-506	-253	0	0	0	0	255
255	0	0	255	765	1020	765	255
255	255	255	255	255	255	255	255

Gx

255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	255
255	0	0	-255	-255	0	255	255
255	0	0	-765	-765	0	765	255
255	0	0	-1020	-1020	0	1020	255
255	0	253	-1020	-1020	0	1020	255
255	0	506	-765	-765	0	765	255
255	255	255	255	255	255	255	255

255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	255
255	0	0	361	806	1020	806	255
255	0	0	806	1082	1020	1082	255
255	0	0	1020	1020	0	1020	255
255	506	357	1020	1020	0	1020	255
255	0	506	806	1082	1020	765	255
255	255	255	255	255	255	255	255

$$\text{mag} = \sqrt{Gx^2 + Gy^2}$$

The resultant smoothed image

Smoothing Spatial filters

The resultant smoothed image after normalization process

$$output_{norm} = \frac{output - output_{min}}{output_{max} - output_{min}} * 255$$

255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	255
255	0	0	361	806	1020	806	255
255	0	0	806	1082	1020	1082	255
255	0	0	1020	1020	0	1020	255
255	506	357	1020	1020	0	1020	255
255	0	506	806	1082	1020	1082	255
255	255	255	255	255	255	255	255

Normalization

255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	255
255	0	0	85	190	240	190	255
255	0	0	190	255	240	255	255
255	0	0	240	240	0	240	255
255	119	84	240	240	0	240	255
255	0	119	190	255	240	255	255
255	255	255	255	255	255	255	255

Threshold

255	255	255	255	255	255	255	255
255	0	0	0	0	0	0	255
255	0	0	255	255	255	255	255
255	0	0	255	255	255	255	255
255	0	0	255	255	0	255	255
255	255	255	255	255	0	255	255
255	0	255	255	255	255	255	255
255	255	255	255	255	255	255	255

The resultant smoothed image after thresholding process

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